Penalty Box: 2D Scattering with Hockey Pucks

(Dated: May 2020)

INTRODUCTION

Collision theory is a subfield of physics involving the study of how projectiles bounce off of targets. The purpose of these experiments - called scattering experiments - can elucidate important information about the target via its interactions with the stream projectiles fired at it. By measuring the distribution of projectiles, one can learn the effective area of interaction.

One of the most famous scattering experiments was performed by Ernest Rutherford and his assistants in 1911, where they were able to determine the structure of the atom by firing α-particles at the nucleus of a gold atom. They were able to measure the distribution of the scattered particles and figure out that the most of an atom's mass is held mostly in the positively-charged nucleus [1].

Since then, scattering experiments have been used to find a number of discoveries in the fields of atomic and subatomic research like nuclear fission, nuclear fusion, and quarks, among many others. Though mostly apt for studies where the target is too small to study with traditional means, scattering experiments can be done with projectiles and targets of any size, like the scattering of pool balls or the scattering of asteroids off a planet or star, or even the scattering of hockey pucks on an ice rink (which will be the focus of this paper). In order to understand how we intend to describe the scattering of projectiles off of hockey pucks, we have to build up some intuition and understanding of the principles behind collision theory as it pertains to scattering. Even though a lot of scattering experiments deal with targets and projectiles that are under the thrall of quantum mechanics, scattering can still be understood through the lens of classical mechanics. The following section will deal with the important concepts of collision theory, most notably the **differential cross section**, which will be vital in understanding the hockey puck problem proposed in the problem section.

BACKGROUND

There are two parameters of vital importance to collision theory: the **scattering angle**, denoted θ, and the **impact parameter**, denoted b. Both of these parameters are shown in Fig. 1. The scattering angle θ is defined as the angle between in the projectile's incident velocity and its outgoing velocity. This scattering angle can range from 0 to π, where $\theta = 0$ relates to the projectile completely missing the target, and $\theta = \pi$ denotes a projectile that hits the target and comes straight back.

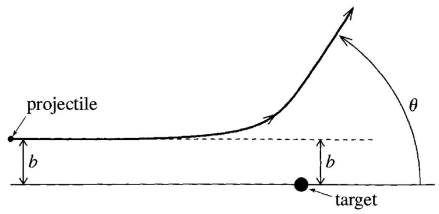

FIG. 1: A simple drawing of a projectile scattering off of a target. The projectile approaches from very far away at a perpendicular distance from the path going through the target's center, which is the impact parameter b. It interacts with the target via the target's contact force and leaves with a velocity displaced from the incident velocity by the scattering angle θ.

The impact parameter b refers to the projectiles perpendicular distance from the path that goes straight through the center of the target. If b is big, there will be little interaction between the target and projectile. If b is small, the projectile force will feel a stronger contact force. Though both of these parameters are vital to scattering theory, it is difficult to directly measure the impact parameter since most experiments deal with atomic and subatomic particles. Because of the difficulty of measuring the impact parameter, a different concept - the **collision cross section** - is a better way of measuring the effective area of influence the target has on the projectile.

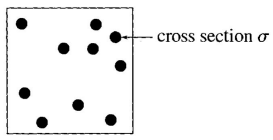

FIG. 2: A basic drawing of a target assembly. The black dots represent the targets as seen from the incident direction, each with a cross-sectional area σ.

If we imagine a beam of projectiles traveling through a collection of targets (like the assembly shown in Fig. 2, we can determine the probability that a projectile will experience an interaction with a target in the assembly. We can find this probability with the **target density** n_{tar} (the number of targets per a given area as seen from the incident direction) the total area of the assembly, and the **cross-sectional area** σ of each target. The total area of all of the targets is thus given by $n_{tar}A\sigma$. We can define the probability of a projectile hitting a target as

it travels through the assembly as

$$(\text{hit probability}) = \frac{\text{target area}}{\text{total area}} = \frac{n_{tar} A\sigma}{A} = n_{tar}\sigma. \quad (1)$$

From this, we can derive an equation for the number of scattered projectiles. The relationship between incident projectiles and scattered projectiles is given by

$$N_{sc} = N_{inc} n_{tar} \sigma \quad (2)$$

where N_{sc} is the number of scattered particles and N_{inc} is the number of incident particles. Since N_{sc}, N_{inc}, and n_{tar} are measurable quantities, we can calculate σ, which we can call the effective area of the target's interaction with the projectile.

The effective area can be described for a number of different processes like capture (when a target absorbs the incident projectile), ionization (when a projectile like an electron ionizes a target by knocking away some electrons), or fission (when a projectile like a neutron collides with a large nucleus and causes it to break apart). Each of these processes can have their own unique σ describing the effective area that a target and projectile can interact in that specific manner. Scattering can be divided further into elastic and inelastic collisions. Elastic collisions are ones in which the internal motions of the target are unchanged by interacting with the projectile, and inelastic collisions are ones where the target's internal motions *do* change.

In cases where a system has multiple interaction methods, the total cross section σ_{tot} is just the sum of the cross sections. For the purposes of this paper, we will be chiefly concerned with the scattering cross section σ_{sc}, since hockey pucks rarely absorb projectiles, experience ionization, or undergo fission.

In Equation 2, we did not pay mind to the direction that the scattered projectiles emerged, only the number of scattered projectiles going in any direction. It can be helpful to take into account these scattered projectiles go to get more information about the target. In order to get this, we need to define the **differential cross section**. Let's consider a scattering experiment such that the only possible interaction is elastic scattering. We can define the scattering direction of a projectile by the polar angles θ and ϕ. It is not difficult to see that attempting to measure every possible combination of polar angles is impossible. We can instead count the scattered projectiles in a cone around a given pair of polar angles (θ, ϕ). In order to do this, we can use the solid angle.

Much like one can define the angle between two lines by measuring the arc length s they define on a circle of radius r, we can define the solid angle of a cone by the area A of a sphere with radius r that the cone intersects. The solid angle is defined as

$$\Delta\Omega = \frac{A}{r^2} \quad (3)$$

where $\Delta\Omega$ is the cone's solid angle, measured in steradians (sr) which has a maximum measurement of 4π sr (just like the maximum measurement of an angle in 2D is 2π rad). For our purposes, we want the narrow cone around a given pair of polar angles with ranges of θ to $\theta + d\theta$ and ϕ to $\phi + d\phi$. The area this cone intersects is $A = r^2 \sin\theta d\theta d\phi$, which gives us the solid angle of our narrow cone:

$$d\Omega = \sin\theta d\theta d\phi. \quad (4)$$

Let's revisit Equation 2, now considering we want to count the number of scattered projectiles into a certain cone. We can rewrite the equation as

$$N_{sc}(\text{into } d\Omega) = N_{inc} n_{tar} d\sigma(\text{into } d\Omega) \quad (5)$$

where $d\sigma(\text{into } d\Omega)$ is the cross section for scattering into the solid angle $d\Omega$. We can define this cross section as

$$d\sigma(\text{into } d\Omega) = \frac{d\sigma}{d\Omega} d\Omega \quad (6)$$

where $d\sigma/d\Omega$ is our **differential scattering cross section**. We can plug this back into Equation 5 to get

$$N_{sc}(\text{into } d\Omega) = N_{inc} n_{tar} \frac{d\sigma}{d\Omega}(\theta, \phi) d\Omega. \quad (7)$$

As we mentioned before, all one needs to do is solve for σ in order to find the total scattering cross section. In this case, we will need to integrate all possible solid angles in order to get N_{sc} in its entirety. Solving for σ in Equation 7 will look like

$$\sigma = \int \frac{d\sigma}{d\Omega}(\theta, \phi) d\Omega = \int_0^{\pi} \sin\theta d\theta \int_0^{2\pi} d\phi \frac{d\sigma}{d\Omega}(\theta, \phi). \quad (8)$$

The differential cross section is going to look different for every situation we encounter. In our hockey puck scenario, we will be considering 2D scattering, but it will prove helpful to look at the 3D case first. Let's consider incident projectiles that approach with impact parameters in a range b to $b + db$ that scatter with angles in the range θ to $\theta + d\theta$. To simplify, let's also assume that the scattering is symmetric about the z-axis so we can consider all ϕ angles simultaneously. This situation is shown in Fig. 3.

The cross-sectional area of annulus representing the approaching incident projectiles is $d\sigma = 2\pi b db$ and the emerging solid angle is $d\Omega = 2\pi \sin\theta d\theta$. The differential cross section for the three dimensional case is

$$\frac{d\sigma}{d\Omega} = \frac{b}{\sin\theta} \left| \frac{db}{d\theta} \right| \quad (9)$$

Knowing this will help us to understand the two-dimensional analogue that we will use in the hockey puck problem, which we will begin discussing now.

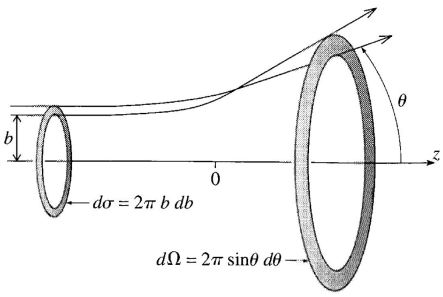

FIG. 3: A representation of incident projectiles approaching a target in a range b to $b + db$ and scattering with angles in a range θ to $\theta + d\theta$. The incident projectiles approaches on an annulus of cross-sectional area of $d\sigma$ and the scattered particles leave within the solid angle $d\Omega$ on the right.

THE PROBLEM

Let's consider a more macroscopic problem, one of the hard-disc of scattering hockey pucks on an ice rink. This problem is similar to traditional three dimensional experiments with subatomic particles but with some simplifying features. For one, we can assume that the scattering will only be in two dimensions and thus won't have to worry about the solid angle of the ice rink. Additionally, the only interaction between the projectiles and the target is elastic scattering where the projectile directly hits the hockey pucks. We don't need to concern ourselves with other forces or interactions like absorption or fission.

With these simplifying assumptions in mind, we first need to find the 2D analogue of the differential cross section. Since we are in two dimensions, we can exchange the solid angle $d\Omega$ for the arc angle $d\theta$. The impact parameter, however, will remain the same. Our differential cross section will instead have the form

$$\frac{d\sigma}{d\theta} \tag{10}$$

where $d\sigma$ is the cross section of the incident particles approaching in a range from b to $b+db$. In 3D, we measured the area of the incident annulus and the solid angle of the emerging ring. In 2D, the area of approach now becomes a length and the emerging range is an arc angle. Consider a beam of projectiles scattering off a hockey puck of some radius. We will need to find the length of the incident range and the arc angle after the scattering occurs.

The length of the small incident range is given by $(b + db) - b$ which is just b. Because the projectile can scatter above and below the primary z axis, the cross-sectional length is

$$d\sigma = 2db$$

to account for both cases. After the projectiles incident in the range scatter above and below the target, the emerge

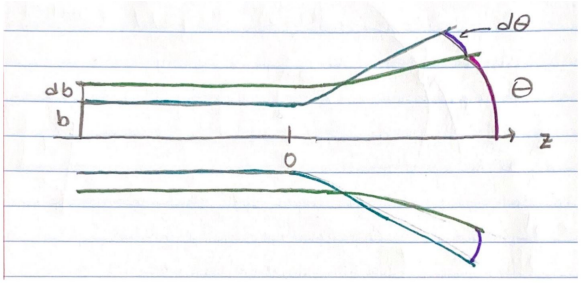

FIG. 4: The 2D analogue of Fig. 3. Instead of an incident area, the 2D analogue features an incident length. The scattered projectiles fire in an arc angle $d\theta$. The differential cross section of this 2D situation is derived below.

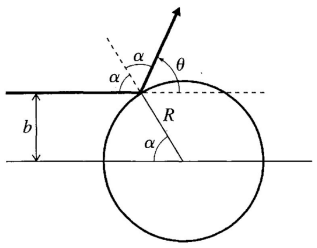

FIG. 5: The scattering of a projectile off a hard disk of radius R. The angle α represents the angle from the z-axis to the point on the disk where the projectile makes contact.

in a narrow angular range from θ to $\theta + d\theta$. This narrow band is just the arc angle which is given by $d\theta$. Thus, our arc angle is

$$d\theta = 2d\theta.$$

Thus, the 2D analogue of the 3D differential cross section is

$$\frac{d\sigma}{d\theta} = \left|\frac{2db}{2d\theta}\right| = \left|\frac{db}{d\theta}\right|. \tag{11}$$

Let's consider the scattering of a projectile of negligibly small radius off of a hard disk of radius R as shown in Fig. 5. In order to find the differential cross section of this scenario, we have to find the impact parameter b in terms of the scattering angle θ, or $b = b(\theta)$. From there, we can differentiate to find $db/d\theta$. Once we get that, we can integrate with respect to θ to get our total cross section.

The projectile has an impact parameter of $b = R\sin\alpha$ which we can see from the geometry of Fig. 5. We can also see that $\theta = \pi - 2\alpha$. In order to find $b(\theta)$, we can solve for α in terms of θ and combine the equations. Since $\alpha = (\pi - \theta)/2$, we can conclude that

$$b = R\sin\frac{\pi - \theta}{2} = R\cos(\theta/2).$$

All we have to do now is differentiate with respect to θ

to get the differential cross section. Doing so gives us

$$\frac{d\sigma}{d\theta} = \left|\frac{db}{d\theta}\right| = \frac{R}{2}\sin(\theta/2) \tag{12}$$

which we can integrate to find the effective cross section of the hockey puck target. When we integrate over all possible scattering angles, we find that

$$\sigma = \int_0^{2\pi} \frac{d\sigma}{d\theta} = \int_0^{2\pi} \frac{R}{2}\sin(\theta/2) = 2R \tag{13}$$

which is just the width of the hockey puck. This makes sense, since the incident projectile will only scatter off the hockey puck if it can actually come in contact with it since there are no additional forces like the Coulomb force that one would see in normal scattering experiments.

Now let's consider the case where our projectile does not have a negligible radius and instead has a radius of twice the hockey puck. What will our effective cross section be now? We will take the impact parameter b to be the distance from the z-axis to the center of the projectile. This situation is shown in Fig. 6.

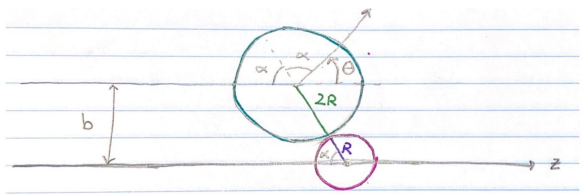

FIG. 6: A hard disk of radius $2R$ scattering off a stationary disk target of radius R. The projectile disk approaches with impact parameter b and scatters with an angle θ.

Finding the cross section is much like the previous scenario. The impact parameter is given by $b = 3R\sin\alpha$ and the scattering angle is the same as before: $\theta = \pi - 2\alpha$. Solving for α again and finding $b(\theta)$ gives us

$$b = 3R\cos(\theta/2).$$

Differentiating with respect to θ gives us

$$\frac{d\sigma}{d\theta} = \left|\frac{db}{d\theta}\right| = \frac{3R}{2}\sin(\theta/2) \tag{14}$$

and integrating over all possible scattering angles gives us

$$\sigma = \int_0^{2\pi} \frac{d\sigma}{d\theta} = \int_0^{2\pi} \frac{3R}{2}\sin(\theta/2) = 6R. \tag{15}$$

This once again makes sense, as the projectile can only scatter off the puck if it touches it. The maximum possible distance where the projectile can still scatter is $3R$, where the projectile is directly on top of the target. This leads to an effective scattering cross section of $6R$, which is the width of the projectile/target combination.

Now let's consider a specific situation. Suppose we have a large ice rink about one 1 kilometer wide with a random arrangement of 2000 hockey pucks pinned to the rink such that none of them hide behind each other. Each of these pucks has a radius of R and our projectiles will be larger pucks of radius $2R$. In this case, $R = 3$ cm. What fraction of the pucks will make it through the assembly without scattering?

We can figure this out by referring back to Equation 5. Since we know σ for the hard disk projectile/target interaction, we can find the fraction of scattered particles by determining the target density n_{tar}. The target density is just the number of targets per length. For this situation, we have 2000 targets per kilometer, or

$$n_{tar} = \frac{2000 \text{ targets}}{1000 \text{ m}} = 2\frac{\text{targets}}{\text{m}}.$$

With the radius $R = 3$ cm $= 0.03$ m, the effective cross section of scattering is $\sigma = 6 * 0.03$ m $= 0.18$ m. We can conclude that the fraction of scattered projectiles is

$$N_{sc} = N_{inc} n_{tar} \sigma = (2 \times 0.18) N_{inc} = 0.36 N_{inc}.$$

This means that the number of projectiles that make it through without scattering at all is just $N_{not} = N_{tot} - N_{sc} = N_{inc} - N_{sc}$, or

$$N_{not} = N_{inc} - 0.36 N_{inc} = 0.64 N_{inc}.$$

The amount of projectiles that make it through the assembly without scattering is 64% of the total incident projectiles. This isn't surprising, as the targets are pretty small, and there are only two of them for every meter in our colossal ice rink. We should expect over half of the projectiles to miss.

We ran a simulation to see if we get the theoretical results. We randomly generated 2000 targets in a one-dimensional range from 0 to 1000. The 1D array ensures that none of the targets are "hiding" behind any others. Additionally, we checked to make sure that none of the pucks overlapped each other. The code for defining the pucks and scattered the projectiles can be found in the Appendix.

We sent 3000 projectiles through the assembly and counted the number of scattered projectiles and the amount of projectiles that missed. In our simulation, we found that 928 projectiles scattered and 2072 projectiles missed. This means that, for our simulations, about 69% missed and about 31% scattered. This is very close to our expected fractions of 64% and 36% for missing and scattering, respectively. One explanation for why our simulated results do not exactly match our expected results is small sample size. In real scattering experiments with subatomic particles, the projectiles come in a beam and at much higher rates. There are tons and tons of incident particles. We, however, only used a relatively small number of projectiles. If we were to fire billions and

billions of pucks (regardless of how unfeasible this would be), we would most likely see results that agree with our theoretical predictions.

APPENDIX

In this Appendix we present the code used to run the simulation. Listing 1 shows the code used to generate the target array and Listing 2 shows the code used to determine the scattered projectiles.

```
r = 0.03
t = 0
pucks = []
while t < 2000:
    puck = nr.rand(1)*1000
    pucks.append(puck)
    x = 0
    for i in pucks:
        if i - 2*r < puck < i + 2*r:
            x = 1
            break
        else:
            x = 0
    if x == 1:
        t = t + 1
    elif x == 0:
        np.delete(pucks,t)
```

Listing 1: Target generation.

We utilized a numpy `while` loop to generate our 2000 target pucks. Each loop creates a puck and appends it to a list of puck positions. Each new puck is checked against all of the previous pucks to see if it lies within a $2R$ distance of an existing puck. If it does, the flag variable `x` is set to `1` to used to mark the puck as invalid. Invalid pucks are deleted and the loop is run again. If the puck doesn't interfere, the loop iterates and moves on to the next puck until there are 2000 targets.

```
u = 0
sc = []
miss = []
while u < 3000:
    proj = nr.rand(1)*1000
    x = 0
    for i in pucks:
        if i - 3*r < proj < i + 3*r:
            x = 1
            break
        else:
            x = 0
    if x == 1:
        sc.append(proj)
    elif x == 0:
        miss.append(proj)
    u = u + 1
```

Listing 2: Scatter testing.

The scatter simulation runs similarly to the target generation. Each iteration of the loop generates a new projectile. If the projectile puck is found to lie within the scattering region (a distance of $3R$ on either side of the center) then it is appended to a list counting the scattered projectiles. If it misses all the targets, it is appended to the missed array. The lengths of these arrays are counted to find the total number of missed projectiles.

[1] E. Rutherford. The Scattering of α and β rays by Matter and the Structure of the Atom. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 21(125):669–688, 1911.